

Supplementary information

Forward dynamics equations

Let $\mathbf{q} = [q_1, q_2]^T$ denote joint angles and $\dot{\mathbf{q}} = [\dot{q}_1, \dot{q}_2]^T$ denote angular velocities. From the Gibbs-Appell formulation, the partial derivative of the Appell acceleration energy with respect to generalized acceleration can be written as:

$$\frac{\partial S}{\partial \ddot{\mathbf{q}}} = M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}),$$

where $M(\mathbf{q})$ is the configuration-dependent mass matrix and $C(\mathbf{q}, \dot{\mathbf{q}})$ contains velocity-dependent inertial terms. Setting this expression equal to the negative gradient of the potential energy gives:

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}) = -G(\mathbf{q}),$$

where $G(\mathbf{q}) = \partial V / \partial \mathbf{q}$. Thus, the ground-truth forward dynamics used for numerical integration were obtained as:

$$\ddot{\mathbf{q}} = -M(\mathbf{q})^{-1}[C(\mathbf{q}, \dot{\mathbf{q}}) + G(\mathbf{q})].$$

For the two-link system, the matrix and vectors were:

$$M(\mathbf{q}) = \begin{bmatrix} l_1^2(m_1 + m_2) + 2l_1l_2m_2 \cos q_2 + l_2^2m_2 & l_2m_2(l_1 \cos q_2 + l_2) \\ l_2m_2(l_1 \cos q_2 + l_2) & l_2^2m_2 \end{bmatrix},$$

$$C(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -l_1l_2m_2 \sin q_2 (2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ l_1l_2m_2 \sin q_2 \dot{q}_1^2 \end{bmatrix},$$

$$G(\mathbf{q}) = \begin{bmatrix} g[l_1(m_1 + m_2) \sin q_1 + l_2m_2 \sin(q_1 + q_2)] \\ gl_2m_2 \sin(q_1 + q_2) \end{bmatrix}.$$

Equivalently, defining $D = m_{11}m_{22} - m_{12}^2 = l_1^2l_2^2m_2(m_1 + m_2 \sin^2 q_2)$, the joint accelerations were:

$$\ddot{q}_1 = \frac{-m_{22}(C_1 + G_1) + m_{12}(C_2 + G_2)}{D},$$

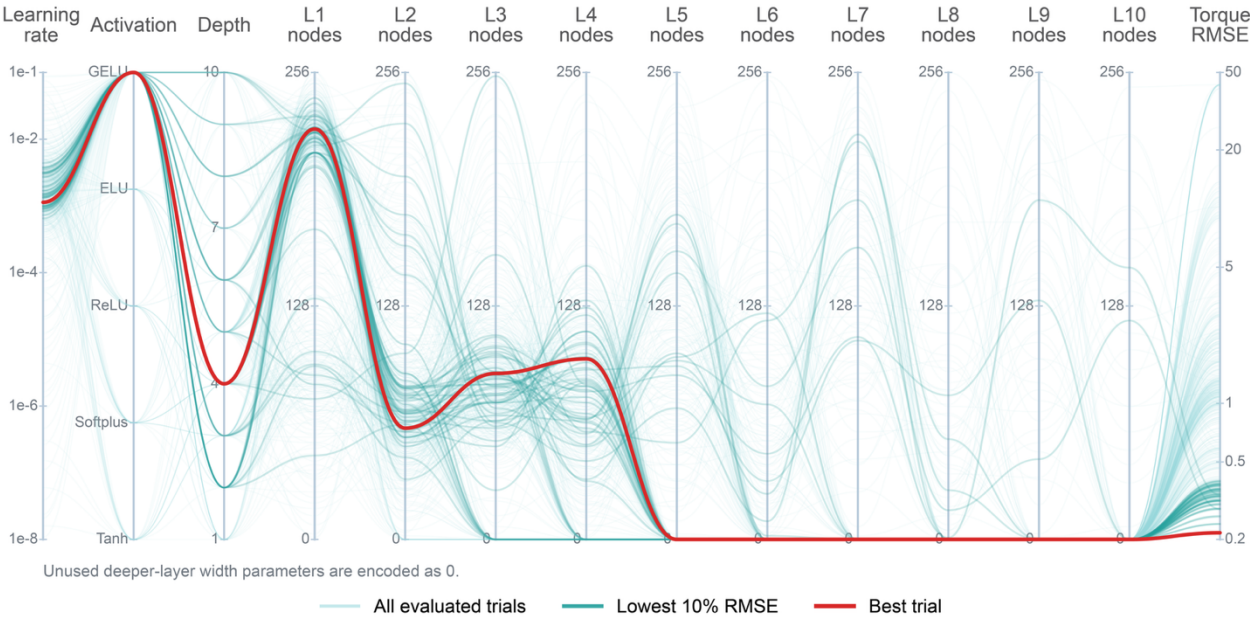
$$\ddot{q}_2 = \frac{m_{12}(C_1 + G_1) - m_{11}(C_2 + G_2)}{D}.$$

These accelerations were then reformulated as the first-order system:

$$\frac{d}{dt} \begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \ddot{q}_1(\mathbf{q}, \dot{\mathbf{q}}) \\ \ddot{q}_2(\mathbf{q}, \dot{\mathbf{q}}) \end{bmatrix},$$

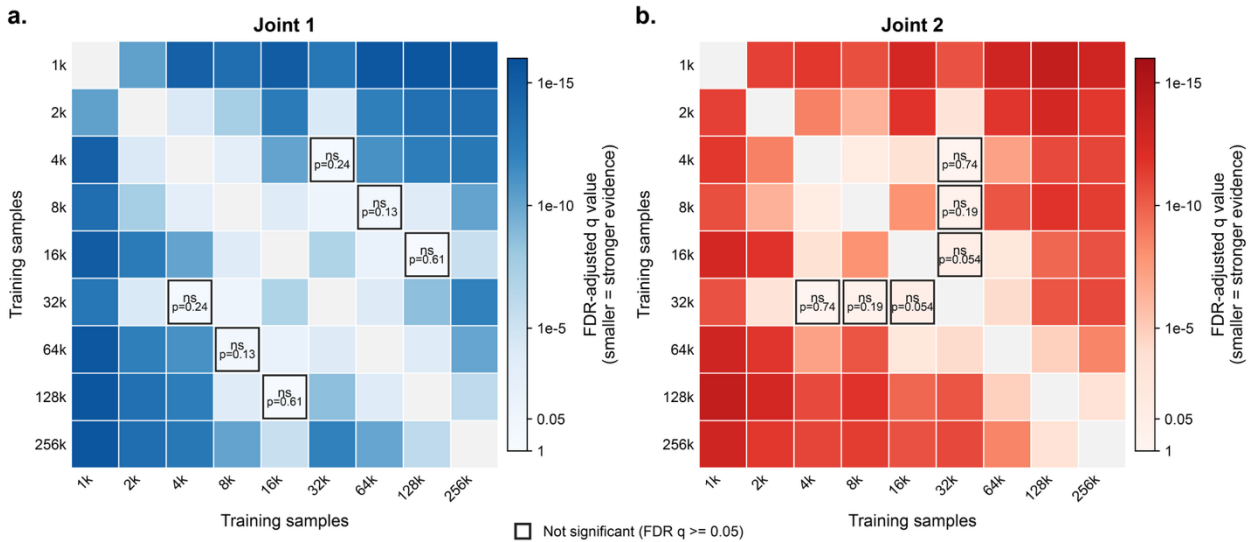
which was integrated using the fixed-step fourth-order Runge-Kutta scheme.

Supplementary Figures and Legends



Supplementary Fig. 1 | Parallel-coordinate representation of hyperparameter optimization.

Each polyline represents one Optuna hyperparameter trial evaluated during the architecture search. Axes show the learning rate, activation function, network depth, hidden-layer widths, and validation torque RMSE. Pale cyan lines show all evaluated trials, teal lines indicate the lowest 10% of trials by RMSE, and the red line marks the best-performing configuration used for subsequent analyses. Unused deeper-layer width parameters in shallower networks are encoded as 0 to display variable-depth architectures on a common axis set.



Supplementary Fig. 2 | Pairwise statistical comparison of inverse-dynamics accuracy across training-set sizes.

a,b, Matrix representation of pairwise comparisons of trajectory-wise torque RMSE distributions for joint 1 (**a**) and joint 2 (**b**) across training-set sizes. Rows and columns denote the dataset sizes being compared. Pairwise differences were evaluated using two-sided Mann-Whitney U tests with Benjamini-Hochberg false discovery rate correction. Cell color reports the FDR-adjusted p value, with darker blue for joint 1 and darker red for joint 2 indicating stronger evidence that the corresponding RMSE distributions differ. Gray diagonal cells represent self-comparisons. Outlined cells labeled ns denote comparisons that were not significant after FDR correction ($p \geq 0.05$).